TG-Analysis: a guided tour of the methods

Companion document to the `tga-analyze` script

This is a tutorial-style walk through the numerical methods behind the `tga-analyze` script. It assumes you know what a TGA trace is and how to read one but does not assume any background in signal processing, nonlinear least squares, or thermal-analysis kinetics. The goal is for the methods to feel obvious by the end, not opaque.

Contents

1 What the instrument gives us, what we want from it

A thermogravimetric analyzer (TGA) heats a sample at a constant rate (commonly $10\,\mathrm{K\,min^{-1}}$) under controlled atmosphere, and records the sample mass W as a function of temperature T. Every chemical event that releases or absorbs mass — water released from a hydrate, CO_2 released from a carbonate, oxygen taken up by a metal, and so on — appears as a step in $W(T)$, sometimes sharp, sometimes sluggish, sometimes piled on top of a neighbor.

What we want from the trace is, for each chemical event:

- where it happens (a characteristic temperature),
- how much mass it accounts for (in % of the starting mass),
- and, when neighboring events overlap, a defensible way to attribute mass to each.

The whole pipeline in this package is dedicated to doing those three things in a way that is reproducible and that flags its own uncertainty when the data are ambiguous.

An aside on what "drift" really is

The chemical mass loss is what we want; everything else is drift. The drift has real physical sources, not just measurement error:

- *Gas buoyancy.* The sample pan displaces a volume of the surrounding gas. As the furnace heats, the gas density falls (roughly as $1/T$ at fixed pressure), so the pan's apparent weight increases by a smooth amount of order 10^{-3} % °C^{-1} over a typical ramp. The instrument can be "blank-corrected" to cancel most of this, but the cancellation is imperfect, especially when sample geometry differs from the blank.
- *Sample-holder expansion.* The crucible and balance arm expand thermally; the center of mass shifts; the apparent weight drifts.
- *Slow parallel processes.* Adsorbed water from atmospheric humidity desorbing over a wide temperature range; trace organic contaminants oxidizing; and so on. These look like drift because they are too slow to register as a discrete event.

The point is that "drift" is a real signal we have to model, not a nuisance we can ignore. It is just slow enough and smooth enough that it does not look like an event.

2 From a noisy mass curve to a clean DTG

The natural quantity for spotting events is not $W(T)$ itself but its derivative, $\mathrm{d}W/\mathrm{d}T$, called the *derivative thermogravimetric* signal or DTG. A flat $W(T)$ corresponds to a zero DTG; an event shows up as a peak in $-\mathrm{d}W/\mathrm{d}T$ (mass loss rate). Peaks are easier to locate, characterize, and separate than steps.

2.1 Why differentiation amplifies noise

A finite-difference estimate of the derivative is

$$\widehat{\mathrm{d}W/\mathrm{d}T}(T_i) = \frac{W(T_{i+1}) - W(T_{i-1})}{T_{i+1} - T_{i-1}}.$$

If W carries noise of standard deviation σ, the numerator contains noise of $\sqrt{2}\,\sigma$, and dividing by the small grid spacing $T_{i+1} - T_{i-1}$ amplifies it. The smaller the spacing, the worse it gets: noise of 0.001 % in W at 0.05 °C spacing becomes noise of about 0.02 % °C^{-1} in $\mathrm{d}W/\mathrm{d}T$, comparable to the weakest events we want to see. Naive numerical differentiation of TGA data is therefore unusable; some kind of smoothing is required, and it must be done in a way that does not destroy the derivative we want to compute.

2.2 Savitzky–Golay: local polynomial smoothing

The Savitzky–Golay (SG) filter is the standard remedy. Around each sample point it fits a polynomial of order p (we use $p = 3$) to the $2k + 1$ nearest neighbors by ordinary least squares, then evaluates that polynomial at the center point to get the smoothed value. The crucial property is that the polynomial coefficients also give the smoothed *derivatives* consistently: instead of differentiating a noisy curve, we read off the analytical derivative of the local fit. The smoothed signal and the smoothed DTG come from the same fit and agree by construction.

To make this concrete: suppose at temperature T_i we take the seven nearest data points $W_{i-3}, W_{i-2}, \ldots, W_{i+3}$ and fit a cubic

$$W(T) \;\approx\; c_0 + c_1(T - T_i) + c_2(T - T_i)^2 + c_3(T - T_i)^3$$

by ordinary least squares. The smoothed value at T_i is just c_0. The smoothed first derivative at T_i is just c_1 — because the derivative of the cubic, evaluated at $T = T_i$, is c_1. The smoothed second derivative is $2c_2$. We get all of these from the same fit; we never differentiate the noisy data, only the smooth polynomial.

We then slide the window over to point T_{i+1} and repeat. Here is the practical magic that makes this fast. Because the least-squares fit is linear in the data, the fitted coefficient c_0 that we report as the smoothed value at T_i is just a fixed weighted sum of the seven raw W values in the window:

$$W_{\text{smoothed}}(T_i) \;=\; c_0 \;=\; \sum_{j=-k}^{+k} h_j\, W_{i+j} \;=\; h_{-3}\, W_{i-3} + h_{-2}\, W_{i-2} + \cdots + h_{+3}\, W_{i+3}.$$

The seven numbers $h_{-3}, h_{-2}, \ldots, h_{+3}$ are fixed: they depend on the half-width k and the polynomial order p, but not on the data values. You can compute them once and store them in a table. The indices $j = -k, \ldots, +k$ are the "offsets" — the positions of the window's points relative to its center, i. On a uniform grid with spacing ΔT, offset j corresponds to the temperature $T_i + j\,\Delta T$.

A weighted sum of a signal at fixed offsets, with weights that do not depend on position, is called a *discrete convolution*. The filter applies the same seven weights at every i, sliding across the whole trace, so smoothing the signal is just one pass of multiply-and-add. We do not re-solve the least-squares problem at each point — the problem was solved once, abstractly, and what remains is mechanical arithmetic. That is why SG is fast.

The same trick gives the first derivative. The fitted coefficient c_1 from the same local cubic is the slope of the smooth polynomial at T_i, which is what we want for the DTG. It is another fixed weighted sum of the same seven W values, just with a different table of weights $h'_{-3}, \ldots, h'_{+3}$:

$$\widehat{\mathrm{d}W/\mathrm{d}T}(T_i) \;=\; c_1 \;=\; \sum_{j=-k}^{+k} h'_j\, W_{i+j}.$$

So the smoothed mass and the smoothed DTG come from the same algorithm with two different precomputed tables of weights. They are guaranteed to be consistent with each other because they were both read off the same local polynomial.

A few intuitions worth carrying:

- A narrow window (k small) follows the data closely and leaves noise behind; the smoothed curve looks essentially identical to the raw curve, and the derivative is still noisy.
- A wide window forces the local polynomial to compromise across many points; noise averages out, but real features — especially sharp peaks — get washed out as well.
- For a given polynomial order, the window width is the only knob; getting it right is the only hard part.

2.3 Choosing the window by residual whiteness

Picking the window by hand is unreliable across runs of different sample sizes or different sample masses, so we let the data choose. The principle: when smoothing is correctly tuned, the residuals $r_i = W_i - W_{\text{smoothed}}(T_i)$ should look like white noise — uncorrelated from one point to the next. If we have smoothed too hard, residuals contain leftover signal, and consecutive residuals are correlated. The simplest test for that correlation is the lag-1 autocorrelation,

$$\rho_1 = \frac{\sum_{i=1}^{n-1}(r_i - \bar{r})(r_{i+1} - \bar{r})}{\sum_{i=1}^{n}(r_i - \bar{r})^2}, \qquad \bar{r} = \frac{1}{n}\sum_{i=1}^{n} r_i,$$

where $\bar{r}$ is the mean residual (the bar denotes a sample average, in the standard statistics convention). Subtracting $\bar{r}$ from each residual removes any constant offset before measuring the correlation, so ρ_1 depends only on whether neighboring residuals move together, not on whether their average happens to be a hair above or below zero. For truly white residuals, $\rho_1 \approx 0$; for over-smoothed residuals, $\rho_1 > 0$. We sweep candidate window widths between 0.3 °C and 5 °C, compute ρ_1 for each, and pick the *largest* window whose $|\rho_1|$ stays below a small threshold (we use 0.05). That is the most aggressive smoothing that has not yet started to eat real signal.

In practice on $10\,\mathrm{K\,min^{-1}}$ runs of $CaCO_3$ powders the procedure picks windows of 0.6 °C to 1.2 °C; for noisier hydrated-cement-paste data it can grow to 4 °C to 5 °C. The chosen width is reported per file.

The summary CSV also reports the *Durbin–Watson* statistic on the residuals,

$$\mathrm{DW} = \frac{\sum_{i=2}^{n}(r_i - r_{i-1})^2}{\sum_{i=1}^{n} r_i^2} \approx 2\,(1 - \rho_1).$$

It carries the same information as ρ_1 but in a more familiar form from classical regression diagnostics: DW ≈ 2 for white residuals, DW → 0 for over-smoothed (positively correlated) residuals, and DW → 4 for residuals that alternate in sign. We report it as a cross-check on ρ_1, not as an additional criterion — the window selection is driven by ρ_1 alone.

3 Finding events

With a clean DTG in hand, events are local maxima of $|\mathrm{d}W/\mathrm{d}T|$. We use `scipy.signal.find_peaks` with two thresholds:

- a *prominence* threshold (default 2 % of the run's maximum $|\mathrm{d}W/\mathrm{d}T|$) to filter out noise peaks, and
- a *minimum separation* (default 20 °C) so a single decomposition with internal wiggles is not double-counted.

For each surviving peak we walk outward in T until $|\mathrm{d}W/\mathrm{d}T|$ drops below a small fraction of the peak amplitude — typically 2 %. We call those crossing temperatures the *event extent*: the range over which we will integrate mass loss for that event. If we walk all the way to a neighboring peak without ever finding a quiet point — common in crowded data like hydrated cement paste — we fall back to the inter-peak DTG minimum, which is the closest local approximation to a baseline that exists.

4 Quantifying mass loss for one event

Once an event is bracketed, we want a number: how much mass was lost during that event? This is harder than it looks, because the instrument records two things superimposed on each other:

- the actual chemical mass loss we care about, and
- a slow "drift" that has nothing to do with the chemistry — buoyancy of the surrounding gas changing with temperature, thermal expansion of the sample holder, slow parallel processes, baseline error of the balance.

Outside an event, the drift is the entire signal — we can measure it directly. *Inside* an event, the chemical signal dwarfs the drift, so the drift is hidden. Quantifying mass loss correctly means estimating, somehow, what the drift would have been under the event. The choices the package offers differ only in how aggressively they try to do that.

4.1 Method 1: naive subtraction

The simplest possible answer is just the smoothed mass at event start minus smoothed mass at event end. This is the `naive` column. It attributes the *entire* mass change in the event extent to the chemistry, ignoring drift. It is often closest to "what the balance shows you".

4.2 Method 2: the ICTAC sloped tangent

The International Confederation for Thermal Analysis and Calorimetry (ICTAC) recommends a tangent construction that is robust to drift without requiring an explicit drift model under the event:

1. Fit a straight line to a quiet region of $W(T)$ *before* the event (the "pre-baseline").
2. Fit a straight line to a quiet region *after* the event (the "post-baseline").
3. Take the tangent line to the smoothed mass curve at the inflection point — the temperature where DTG is at its peak, i.e. where mass is changing fastest.
4. The onset temperature T_{on} is the intersection of the pre-baseline with the inflection tangent. The endset T_{end} is the intersection of the inflection tangent with the post-baseline.
5. Mass loss is the pre-baseline value at T_{on} minus the post-baseline value at T_{end}.

Figure 1 shows the geometry.

This is the `tangent` column. The geometric idea is that we are finding the two points where the steepest part of the curve "intercepts" the extrapolated baselines on either side. It corrects for drift because the pre- and post-baselines, being sloped, account for drift on each side; the question of *how* drift transitions between them inside the event never arises because the construction only evaluates each baseline at its own end.

Many commercial instruments use a horizontal-tangent version of this construction (the baselines are forced flat). When the trace has any drift, the horizontal version reduces algebraically to the naive subtraction — so the sloped version is what you should be reporting in a paper.

4.3 Method 3: DTG-area with an under-event baseline

If we want to integrate the DTG over the event extent, we have to choose a baseline curve $B(T)$ that lives *inside* the event — the curve we would have seen with no chemistry. The corrected mass loss is then

$$\Delta W = \int_{T_{\mathrm{start}}}^{T_{\mathrm{end}}} \big[\, B'(T) - W'(T) \,\big] \, \mathrm{d}T,$$

where $W'(T)$ is the smoothed DTG and $B'(T) = \mathrm{d}B/\mathrm{d}T$ is the drift contribution to the DTG. The integral is computed numerically (trapezoidal rule) on the uniform T grid. The package reports three choices for $B'(T)$, in increasing order of physical sophistication.

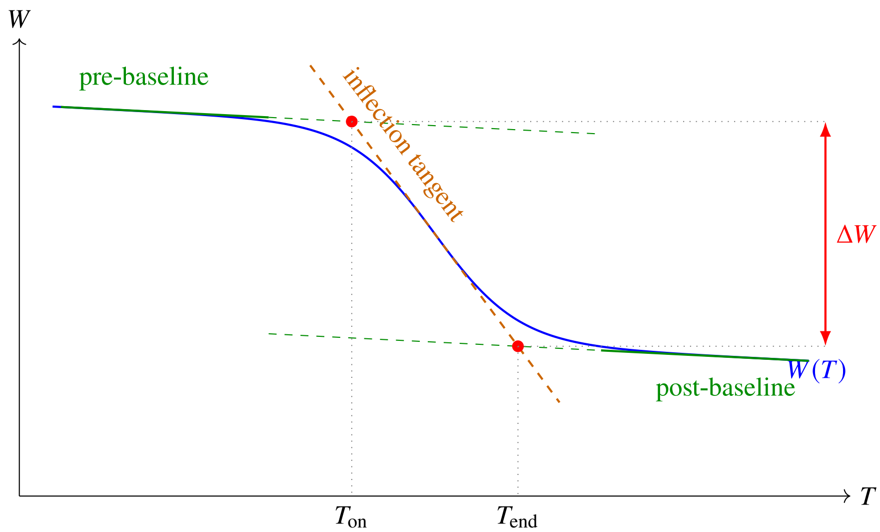

Figure 1: ICTAC sloped-tangent construction. Solid green segments are the pre- and post-baseline linear fits to the quiet regions on either side of the event; the dashed extensions show those baselines extrapolated through the event. The orange dashed line is the tangent to the mass curve at the inflection point (where DTG is largest in magnitude). The intersections with the pre- and post-baselines define T_{on} and T_{end}. Mass loss ΔW is the pre-baseline value at T_{on} minus the post-baseline value at T_{end}. The under-event drift never has to be explicitly modeled because each baseline is only evaluated at its own end.

Constant slope (`dtg_area_constant`). Hold the drift slope fixed at its pre-event value: $B'(T) = a_{\text{pre}}$, where a_{pre} is the slope of the pre-baseline. Interpretation: "the drift we saw before the event just keeps going." Justified when post-event drift is poorly characterized, or when one trusts only the pre-event slope.

Linear blend (`dtg_area_linear`). Interpolate linearly in T between the pre- and post-event slopes:

$$B'(T) \;=\; (1-\alpha_T)\,a_{\text{pre}} + \alpha_T\,a_{\text{post}}, \qquad \alpha_T \;=\; \frac{T - T_{\text{start}}}{T_{\text{end}} - T_{\text{start}}}.$$

Justified when drift varies smoothly with T and is uncoupled from the reaction itself.

Extent-of-reaction blend (`dtg_area_alpha`). Blend the slopes by how much of the reaction has happened so far, rather than by where in the temperature interval we are:

$$B'(T) \;=\; (1-\alpha(T))\,a_{\text{pre}} + \alpha(T)\,a_{\text{post}}, \qquad \alpha(T) \;=\; \frac{W(T_{\text{start}}) - W(T)}{W(T_{\text{start}}) - W(T_{\text{end}})}.$$

For a peaked DTG the cumulative fraction $\alpha(T)$ jumps from near zero to near one over a narrow temperature range around the peak, so the slope transition concentrates there instead of spreading uniformly. This is the Borchardt–Daniels construction recommended by the ICTAC kinetics committee. It is self-consistent when the drift is *tied to the state of the sample* — a sample that has lost 44 % of its mass has different buoyancy and geometry than the intact one, so the drift contribution genuinely should change as the reaction proceeds.

Figure 2 contrasts α_T (linear in T) with $\alpha(T)$ (extent of reaction). The two functions both run from 0 to 1 across the event, but the extent-of-reaction version is a sigmoid concentrated near the inflection. The corresponding under-event baseline slopes therefore transition at different temperatures.

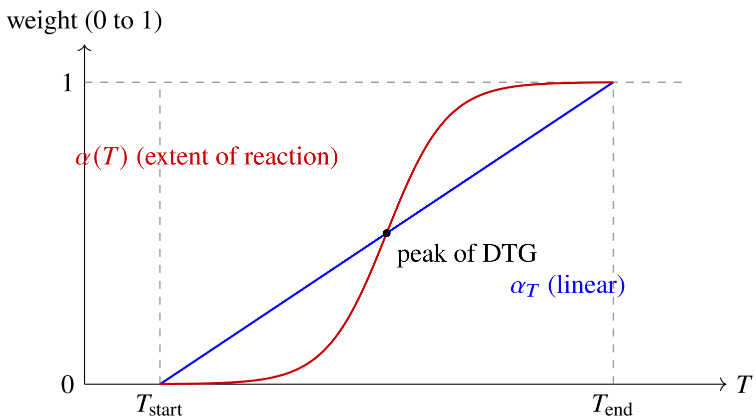

Figure 2: The two weighting functions used in the linear and alpha baseline modes. Linear (α_T) rises uniformly with temperature; extent-of-reaction ($\alpha(T)$) follows the cumulative mass loss, so it jumps sharply through the inflection (where the DTG is peaked) and plateaus on either side. The under-event drift therefore transitions its slope at different temperatures under the two conventions, which is why the corresponding integrated mass losses can differ.

4.4 An identity worth keeping in mind

A subtle point: although the constant and linear baselines look quite different pointwise, they give the *same integrated* mass loss if you define "constant" as the mean of a_{pre} and a_{post}. The reason is just calculus — the integral of a linear function over an interval equals its midpoint times the width:

$$\int_{T_{\text{start}}}^{T_{\text{end}}} \left[(1-\alpha_T)\,a_{\text{pre}} + \alpha_T\,a_{\text{post}}\right] \mathrm{d}T \;=\; \tfrac{1}{2}(a_{\text{pre}} + a_{\text{post}})\,(T_{\text{end}} - T_{\text{start}}).$$

To see this explicitly: substitute $u = (T - T_{\text{start}})/(T_{\text{end}} - T_{\text{start}})$, so $\alpha_T = u$ and $\mathrm{d}T = (T_{\text{end}} - T_{\text{start}})\,\mathrm{d}u$. The integral becomes $(T_{\text{end}} - T_{\text{start}}) \int_0^1 \left[(1-u)a_{\text{pre}} + u\,a_{\text{post}}\right]\mathrm{d}u = (T_{\text{end}} - T_{\text{start}}) \cdot \tfrac{1}{2}(a_{\text{pre}} + a_{\text{post}})$. The linear-blend baseline and the mean-slope constant baseline differ pointwise at every T except the midpoint, and yet their integrals agree exactly. The pointwise difference cancels by symmetry of the trapezoid.

For that reason, this package's `dtg_area_constant` uses a_{pre} alone (not the mean), so the constant and linear columns are genuinely distinct numbers. The alpha column differs from both only when a_{pre} and a_{post} differ appreciably and the DTG is asymmetric — which is most of the time on real data.

5 When events overlap: deconvolution

Everything above assumes that each event has quiet baseline on both sides. In a hydrated cement paste below 200 °C, the calcium-silicate-hydrate, ettringite, and aluminate hydrates dehydrate on top of each other; there is no quiet region between them. The peak detector picks up one feature where there are really three or four, the integration extent sweeps up neighboring chemistry, and the reported number is structurally wrong, not just imprecise.

The fix is to model the region as a sum of asymmetric peak shapes and fit them to the baseline-corrected DTG.

5.1 The Frazer–Suzuki peak shape

The standard peak function in thermal-analysis kinetics is the Frazer–Suzuki form

$$y(T; A, T_0, W, b) \;=\; A\,\exp\left[-\frac{\ln 2}{b^2}\,\ln^2\!\left(1+\frac{2b\,(T-T_0)}{W}\right)\right].$$

Its four parameters have direct interpretations:

- A is the peak amplitude (in % °C^{-1}).
- T_0 is the peak temperature.
- W is the full width at half maximum (exactly so in the symmetric limit $b \to 0$).
- b is an asymmetry. With $b = 0$ the function is a Gaussian. Positive b gives a right tail; negative b gives a left tail. Real thermal decompositions almost always have left tails (a slow ramp-up of reaction rate followed by a sharp fall), so fitted values are typically $-0.9 \lesssim b < 0$.

The function vanishes outside the temperature range where its log argument is positive. In mathematical language this property is called *compact support*: the set of temperatures where the function is non-zero is bounded (it fits inside a finite interval), and *outside* that interval the function is exactly zero, not just very small. This is in contrast to Gaussians and similar bell shapes, which are technically non-zero everywhere — they decay exponentially but never quite reach zero. For deconvolution the distinction matters in practice: a Frazer–Suzuki peak placed at T_0 contributes exactly zero at temperatures well beyond its own extent, so a fit on the carbonate region around 700 °C cannot be contaminated by the residual exponential tail of a peak the fit also placed at, say, 200 °C. Each peak influences only the points within its own bounded range.

5.2 Fitting a sum of peaks

If a region $[T_{\mathrm{lo}}, T_{\mathrm{hi}}]$ is believed to contain N overlapping events, we model the baseline-corrected DTG there as

$$y_{\mathrm{model}}(T) \;=\; \sum_{i=1}^{N} y(T; A_i, T_{0,i}, W_i, b_i)$$

and fit the $4N$ parameters by nonlinear least squares (`scipy.optimize.curve_fit`). Initial guesses for $T_{0,i}$ come from a peak detector on y_{model} itself; widths default to a fraction of the region width; asymmetries start at zero. Bounds keep $A \geq 0$, $W \geq 1$ °C, and $|b| \leq 0.95$ to avoid degenerate fits.

The mass loss attributed to peak i is its integral, computed numerically on the same T grid. The peak integrals sum to the total integrated mass loss for the region under the chosen baseline.

5.3 The region must extend into quiescent baseline flanks

By far the most common deconvolution mistake is choosing a region that starts or ends inside the tail of a real event. We use the term *quiescent baseline flank* for the interval of temperature on either side of the fitted region in which the mass signal is dominated by slow, near-linear drift and no active decomposition; this is the substrate on which a defensible pre- or post-event baseline can be fitted. The package fits the pre-baseline to a 20 °C-wide window just below T_{lo} and the post-baseline to a 20 °C-wide window just above T_{hi}. If either of those windows is contaminated by ongoing chemistry — i.e., if it does not lie in a quiescent flank — the baseline slope is wrong, the corrected DTG is wrong, and the deconvolved per-peak masses are nonsense, even though the fit converged.

A worked example. On a clean calcite (theoretical mass loss 43.97 %), a deconvolution with $T_{\mathrm{lo}} = 700$ °C, which sits well inside the rising edge of the decarbonation event, gives per-peak mass losses of approximately

21 %, 27 % and 27 % under the three baseline modes. Widening to $T_{\mathrm{lo}} = 540\,^{\circ}\mathrm{C}$ — which puts the leading window into a genuinely quiescent flank — gives 43.6 %, 43.7 % and 43.7 %: a 0.3 % spread, agreeing with theory. The fit converged in both cases. The only difference is whether the leading window saw chemistry or not.

6 What deconvolution cannot do

It is important to be honest about the resolution limit of any single-ramp TGA experiment. Deconvolution is not magic: it imposes a parameterized peak shape on the data and finds the best-fit parameters, but it cannot extract information that is not in the trace.

Two events that overlap sufficiently — such that no shoulder or inflection is visible in the DTG — are *mathematically unresolvable* from one ramp. Many different decompositions of the total signal into two peaks fit equally well; the parameters trade off against one another, and the result depends mostly on the initial guess. The fit reports tiny residuals and nonsense uncertainties; if you re-ramp the same sample at a different heating rate, you get a different "deconvolution" that fits equally well.

Honest options when this happens are:

- *Multi-rate (isoconversional) experiments.* Ramp the same sample at, e.g., $5\,\mathrm{K\,min^{-1}}$, $10\,\mathrm{K\,min^{-1}}$, $20\,\mathrm{K\,min^{-1}}$ and $40\,\mathrm{K\,min^{-1}}$, and analyze the apparent activation energy as a function of α. Events with different kinetic signatures separate in the multi-rate data even if they overlap at any single rate. This is the approach the ICTAC kinetics committee recommends.
- *Coupled techniques.* TGA–MS (mass spectrometer on the evolved gas) or TGA–FTIR identifies *what* is being released, so you can attribute mass to chemistry directly rather than guessing from the temperature alone.
- *Sample preparation.* Sometimes the right answer is not better analysis but a different experiment — e.g., washing away the carbonate phase before the ramp to isolate the portlandite peak.

The package will happily deconvolve any region you give it, with any number of peaks you ask for. It is your job to confirm that the data actually contain shoulders that justify the number of peaks — the diagnostic PNG is your friend here. If the corrected DTG in the region is genuinely a single asymmetric bump, fitting two Frazer–Suzuki peaks to it will give you two peaks with arbitrary parameter values that sum to the right shape but mean nothing individually.

7 A practical caveat: $CaCO_3$ polymorphs

If your samples are $CaCO_3$ powders or contain $CaCO_3$ phases, a chemistry caveat worth knowing: the three common $CaCO_3$ polymorphs — aragonite, vaterite, and calcite — do *not* have distinguishable thermal decomposition signatures in TGA. Both aragonite and vaterite undergo a solid-state polymorphic transformation to calcite well before $CaCO_3 \rightarrow CaO + CO_2$:

- Vaterite → calcite, roughly 320 °C to 460 °C.
- Aragonite → calcite, roughly 380 °C to 490 °C.
- $CaCO_3 \rightarrow CaO + CO_2$ (calcite), roughly 600 °C to 800 °C, depending on crystallinity, partial pressure of CO_2, and heating rate.

Whatever polymorph you put in the pan, by the time decarbonation happens the sample is calcite. The decarbonation DTG cannot, in principle, distinguish "calcite that has always been calcite" from "aragonite that became calcite at 430 °C." Coupled techniques — room-temperature XRD, FT-Raman, or FTIR — are required for polymorph identification. This has been documented since Passe-Coutrin *et al.* (1992) and is the explicit subject of Kontoyannis & Vagenas (*Analyst* **125**, 251, 2000), the standard reference.

The polymorph transformations themselves *are* visible in DSC (they are weakly exothermic), and they show up as small features in DTG if the polymorphic transition is accompanied by water release from a hydrate carbonate, but the 700 °C decarbonation peak by itself is silent on which polymorph you started with.

8 Method agreement as a diagnostic

Putting all of this together, the workflow the package supports is:

1. Run the detection and the five mass-loss columns.
2. For each event, look at the three DTG-area columns (`constant`, `linear`, `alpha`). If they agree to within a couple of percent and lie close to `tangent`, the answer is trustworthy — the event is cleanly isolated and any of the methods would do.
3. If the three DTG-area values *disagree*, or if any of them differs greatly from `tangent`, treat the number as suspect. The event extent is being polluted by overlapping chemistry, and no baseline choice can fix it; the integration limits themselves are wrong.
4. In that case, run a deconvolution over a region wide enough to include the event *and* 20 °C of quiescent baseline flank on each side, with N peaks chosen to match what the DTG plot suggests.
5. Confirm that the per-peak masses agree across the three baseline modes. If they do, those numbers are trustworthy. If they still disagree, widen the region or reconsider N.

The point is that the package is not telling you what the answer is. It is telling you when you can trust the answer it computed, and when you need to do more work. The agreement (or disagreement) of three methods that ought to give the same answer under good conditions is itself the validation signal — a useful guard against an over-confident single number from a single method.

Further reading

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